Marketing Campaign Analysis

This notebook analyzes `Marketing_Campaign_Data.csv` to answer the client question:

Which campaign and channel combination should we focus on to increase sales to new customers, and why?

- Campaign A: informal and conversational tone
- Campaign B: formal and corporative tone

The workflow below loads the dataset, compares new-customer performance by campaign and channel, builds visualizations, and exports an HTML dashboard.

```
from pathlib import Path

import pandas as pd
import plotly.express as px
import plotly.graph_objects as go
from plotly.subplots import make_subplots

DATA_PATH = Path(r"c:\Users\davie\OneDrive\Documentos\Data Analyst\
Marketing_Campaign_Data.csv")
OUTPUT_HTML = Path(r"c:\Users\davie\OneDrive\Documentos\Data Analyst\
marketing_campaign_dashboard_from_notebook.html")

# Load and standardize the source data.
df = pd.read_csv(DATA_PATH)
df.columns = [column.strip() for column in df.columns]

numeric_columns = [
    "Converted (1=yes, 0=no)",
    "Time on Site (seconds)",
    "Sales ($)",
]
for column in numeric_columns:
    df[column] = pd.to_numeric(df[column], errors="coerce")

new_customers = df[df["Customer Type"].eq("New")].copy()
new_customers.head()
```

```
   Interaction ID Campaign Type    Channel Customer Type  \
1       INT000002             B      Email           New
2       INT000003             A      Email           New
12      INT000013             B      Email           New
13      INT000014             B  Instagram           New
15      INT000016             B  Instagram           New

    Converted (1=yes, 0=no)  Time on Site (seconds)  Sales ($)
```

```
1                          1              168.161435  61.085973
2                          0              146.636771   0.000000
12                         1              215.246637  43.438914
13                         1              145.739910  34.841629
15                         0              117.937220   0.000000
```

```
overall_kpis = {
    "new_customer_rows": int(len(new_customers)),
    "new_customer_sales": float(new_customers["Sales ($)"].sum()),
    "new_customer_conversion_rate": float(new_customers["Converted
(1=yes, 0=no)"].mean()),
}

combo_summary = (
    new_customers.groupby(["Campaign Type", "Channel"],
as_index=False)
    .agg(
        leads=("Interaction ID", "count"),
        conversions=("Converted (1=yes, 0=no)", "sum"),
        conversion_rate=("Converted (1=yes, 0=no)", "mean"),
        total_sales=("Sales ($)", "sum"),
        sales_per_lead=("Sales ($)", "mean"),
        avg_time_on_site=("Time on Site (seconds)", "mean"),
    )
)
combo_summary["sales_share"] = combo_summary["total_sales"] /
combo_summary["total_sales"].sum()
combo_summary["focus_score"] = (
    0.55 * (combo_summary["total_sales"] /
combo_summary["total_sales"].mean())
    + 0.45 * (combo_summary["sales_per_lead"] /
combo_summary["sales_per_lead"].mean())
)
combo_summary = combo_summary.sort_values(
    ["focus_score", "total_sales", "conversion_rate"],
    ascending=[False, False, False],
).reset_index(drop=True)

campaign_summary = (
    new_customers.groupby("Campaign Type", as_index=False)
    .agg(
        leads=("Interaction ID", "count"),
        conversions=("Converted (1=yes, 0=no)", "sum"),
        conversion_rate=("Converted (1=yes, 0=no)", "mean"),
        total_sales=("Sales ($)", "sum"),
        sales_per_lead=("Sales ($)", "mean"),
    )
    .sort_values("total_sales", ascending=False)
)
```

```
channel_summary = (
    new_customers.groupby("Channel", as_index=False)
    .agg(
        leads=("Interaction ID", "count"),
        conversions=("Converted (1=yes, 0=no)", "sum"),
        conversion_rate=("Converted (1=yes, 0=no)", "mean"),
        total_sales=("Sales ($)", "sum"),
        sales_per_lead=("Sales ($)", "mean"),
    )
    .sort_values("total_sales", ascending=False)
)

recommended_combo = combo_summary.iloc[0]
recommended_combo.to_frame(name="value")
```

```
                          value
Campaign Type                 B
Channel                   Email
leads                      9700
conversions                4673
conversion_rate        0.481753
total_sales       225285.520362
sales_per_lead        23.225311
avg_time_on_site     134.545097
sales_share              0.2279
focus_score            1.202454
```

Recommendation

Focus on Campaign B through Email.

Why this combination wins for new customers:

- It generates the highest total sales in the dataset.
- It converts above the overall new-customer average.
- It combines strong efficiency with enough lead volume to scale.
- Campaign B, the more formal and corporative tone, outperforms Campaign A on total new-customer sales across all channels.

```
display_columns = [
    "Campaign Type",
    "Channel",
    "leads",
    "conversions",
    "conversion_rate",
    "total_sales",
    "sales_per_lead",
    "sales_share",
]
```

```
styled_combo_summary = combo_summary[display_columns].copy()
styled_combo_summary["conversion_rate"] =
styled_combo_summary["conversion_rate"].map(lambda value:
f"{value:.2%}")
styled_combo_summary["total_sales"] =
styled_combo_summary["total_sales"].map(lambda value: f"$
{value:,.2f}")
styled_combo_summary["sales_per_lead"] =
styled_combo_summary["sales_per_lead"].map(lambda value: f"$
{value:,.2f}")
styled_combo_summary["sales_share"] =
styled_combo_summary["sales_share"].map(lambda value: f"{value:.2%}")

print("Overall new-customer KPIs")
print(f"Rows: {overall_kpis['new_customer_rows']:,}")
print(f"Sales: ${overall_kpis['new_customer_sales']:,.2f}")
print(f"Conversion rate:
{overall_kpis['new_customer_conversion_rate']:.2%}")
print()
print("Ranked campaign-channel combinations")
styled_combo_summary
```

```
Overall new-customer KPIs
Rows: 42,597
Sales: $988,528.51
Conversion rate: 47.86%

Ranked campaign-channel combinations

  Campaign Type     Channel  leads  conversions conversion_rate
total_sales  \
0             B       Email   9700         4673          48.18%
$225,285.52
1             A       Email   7683         3628          47.22%
$177,366.97
2             B   Instagram   7100         3429          48.30%
$166,722.17
3             B  Web Banner   7110         3387          47.64%
$164,125.79
4             A   Instagram   5456         2658          48.72%
$127,814.93
5             A  Web Banner   5548         2614          47.12%
$127,213.12

  sales_per_lead sales_share
0         $23.23      22.79%
1         $23.09      17.94%
2         $23.48      16.87%
3         $23.08      16.60%
```

```
4          $23.43      12.93%
5          $22.93      12.87%
```

```
color_map = {"A": "#c96b59", "B": "#2f7f7b"}

top_order = combo_summary[["Campaign Type", "Channel"]].copy()
top_order["label"] = top_order["Campaign Type"] + " - " +
top_order["Channel"]
combo_chart_data = combo_summary.copy()
combo_chart_data["label"] = combo_chart_data["Campaign Type"] + " - "
+ combo_chart_data["Channel"]
combo_chart_data["tone"] = combo_chart_data["Campaign Type"].map({
    "A": "Informal / conversational",
    "B": "Formal / corporative",
})

bar_fig = px.bar(
    combo_chart_data,
    x="label",
    y="total_sales",
    color="Campaign Type",
    text="total_sales",
    color_discrete_map=color_map,
    category_orders={"label": top_order["label"].tolist()},
    hover_data={
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        "tone": True,
        "leads": ':,',
        "conversion_rate": ':.2%',
        "sales_per_lead": ':$.2f',
        "total_sales": ':$.2f',
        "label": False,
    },
    title="New-customer sales by campaign and channel",
)
bar_fig.update_traces(texttemplate="$%{y:,.0f}",
textposition="outside")
bar_fig.update_layout(
    xaxis_title="Campaign and channel",
    yaxis_title="Sales ($)",
    plot_bgcolor="white",
    paper_bgcolor="white",
    legend_title_text="Campaign",
)
bar_fig.show()

scatter_fig = px.scatter(
    combo_chart_data,
    x="conversion_rate",
    y="sales_per_lead",
```

```
    size="leads",
    color="Campaign Type",
    color_discrete_map=color_map,
    hover_name="label",
    hover_data={
        "tone": True,
        "total_sales": ':$.2f',
        "conversion_rate": ':.2%',
        "sales_per_lead": ':$.2f',
        "leads": ':,',
    },
    title="Scale versus efficiency for new-customer combinations",
)
scatter_fig.update_layout(
    xaxis_title="Conversion rate",
    yaxis_title="Sales per lead ($)",
    plot_bgcolor="white",
    paper_bgcolor="white",
    legend_title_text="Campaign",
)
scatter_fig.show()
```


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```
comparison = (
    combo_summary.pivot(index="Channel", columns="Campaign Type",
values=["total_sales", "conversion_rate"])
    .sort_index()
)
comparison_table = pd.DataFrame({
    "Channel": comparison.index,
    "B_minus_A_sales": comparison[("total_sales", "B")] -
comparison[("total_sales", "A")],
    "B_minus_A_conversion_pp": (comparison[("conversion_rate", "B")] -
comparison[("conversion_rate", "A")]) * 100,
}).reset_index(drop=True)
comparison_table

comparison_fig = make_subplots(specs=[[{"secondary_y": True}]])
comparison_fig.add_trace(
    go.Bar(
        x=comparison_table["Channel"],
        y=comparison_table["B_minus_A_sales"],
        name="Revenue lift from B vs A",
        marker_color="#20364f",
        hovertemplate="%{x}<br>Revenue lift:
$%{y:,.2f}<extra></extra>",
    ),
    secondary_y=False,
)
comparison_fig.add_trace(
    go.Scatter(
        x=comparison_table["Channel"],
        y=comparison_table["B_minus_A_conversion_pp"],
        name="Conversion delta (pp)",
        mode="lines+markers+text",
        text=[f"{value:+.2f} pp" for value in
comparison_table["B_minus_A_conversion_pp"]],
        textposition="top center",
        line=dict(color="#ca8f2c", width=3),
        marker=dict(size=9),
        hovertemplate="%{x}<br>Conversion delta: %{y:.2f}
pp<extra></extra>",
    ),
    secondary_y=True,
)
comparison_fig.update_layout(
    title="Campaign B uplift versus Campaign A by channel",
    plot_bgcolor="white",
    paper_bgcolor="white",
    legend=dict(orientation="h", yanchor="bottom", y=1.02,
```

```
xanchor="left", x=0),
)
comparison_fig.update_yaxes(title_text="Revenue lift ($)",
secondary_y=False)
comparison_fig.update_yaxes(title_text="Conversion delta (pp)",
secondary_y=True)
comparison_fig.show()
```


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dashboard_fig = make_subplots(
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{"type": "table"}]],
    subplot_titles=(
        "Sales by campaign and channel",
        "Scale versus efficiency",
        "Campaign B uplift versus A",
        "Ranked combinations",
    ),
    vertical_spacing=0.14,
)

for campaign in ["A", "B"]:
    campaign_slice = combo_chart_data[combo_chart_data["Campaign
Type"] == campaign]
    dashboard_fig.add_trace(
        go.Bar(
            x=campaign_slice["label"],
            y=campaign_slice["total_sales"],
            name=f"Campaign {campaign}",
            marker_color=color_map[campaign],
            showlegend=True,
        ),
        row=1,
        col=1,
    )

dashboard_fig.add_trace(
    go.Scatter(
        x=combo_chart_data["conversion_rate"],
        y=combo_chart_data["sales_per_lead"],
        mode="markers+text",
        text=combo_chart_data["label"],
        textposition="top center",
        marker=dict(
            size=(combo_chart_data["leads"] / 180).round(1),
            color=combo_chart_data["Campaign Type"].map(color_map),
            sizemode="diameter",
            line=dict(color="white", width=1),
        ),
        name="Combinations",
        showlegend=False,
    ),
```

```
    row=1,
    col=2,
)

dashboard_fig.add_trace(
    go.Bar(
        x=comparison_table["Channel"],
        y=comparison_table["B_minus_A_sales"],
        marker_color="#20364f",
        name="Revenue lift",
        showlegend=False,
    ),
    row=2,
    col=1,
)

dashboard_fig.add_trace(
    go.Table(
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            font=dict(color="white", size=12),
            align="left",
        ),
        cells=dict(
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                combo_summary["total_sales"].map(lambda value: f"$
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            ],
            fill_color="#f8f5ef",
            align="left",
            height=28,
        ),
    ),
    row=2,
    col=2,
)

dashboard_fig.update_layout(
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    title_text="Marketing Campaign Dashboard: New-Customer
Acquisition",
```

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    plot_bgcolor="white",
    paper_bgcolor="white",
)

dashboard_fig.update_xaxes(title_text="Campaign and channel", row=1,
col=1)
dashboard_fig.update_yaxes(title_text="Sales ($)", row=1, col=1)
dashboard_fig.update_xaxes(title_text="Conversion rate",
tickformat=".0%", row=1, col=2)
dashboard_fig.update_yaxes(title_text="Sales per lead ($)", row=1,
col=2)
dashboard_fig.update_xaxes(title_text="Channel", row=2, col=1)
dashboard_fig.update_yaxes(title_text="Revenue lift ($)", row=2,
col=1)

dashboard_fig.show()
dashboard_fig.write_html(str(OUTPUT_HTML), include_plotlyjs="cdn")
print(f"Dashboard exported to: {OUTPUT_HTML}")
```


```
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- Web Banner"],"xaxis":"x","y":
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```

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[0.5700000000000001,1],"title":{"text":"Sales ($)"}},"yaxis2":
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```
per lead ($)"}},"yaxis3":{"anchor":"x3","domain":[0,0.43],"title":
{"text":"Revenue lift ($)"}}}}

Dashboard exported to: c:\Users\davie\OneDrive\Documentos\Data
Analyst\marketing_campaign_dashboard_from_notebook.html
```

Dashboard code explained

The dashboard HTML is organized in three layers:

1. **Head section**: loads the fonts, Plotly library, and all CSS styles used by the dashboard layout.
2. **Body section**: defines the page structure, including the hero recommendation block, KPI cards, chart containers, and the ranking table.
3. **Script section**: stores the dashboard data in JavaScript arrays and uses `Plotly.newPlot(...)` to render each chart.

How to read the code:

- The CSS variables in `:root` control the dashboard palette and visual style.
- The `comboData` array contains the ranked campaign and channel combinations.
- The `deltaData` array stores the Campaign B versus Campaign A uplift by channel.
- The `salesByCombo`, `efficiencyPlot`, and `toneDelta` charts are each created with Plotly using those arrays.
- The final table is built dynamically by looping through `comboData` and inserting rows into the table body.

The next cell loads the exact source code from `marketing_campaign_dashboard.html` and displays it directly in the notebook.

```
from IPython.display import Code, display

DASHBOARD_SOURCE_PATH = Path(r"c:\Users\davie\OneDrive\Documentos\Data
Analyst\marketing_campaign_dashboard.html")
dashboard_html_code = DASHBOARD_SOURCE_PATH.read_text(encoding="utf-
8")

print(f"Dashboard source loaded from: {DASHBOARD_SOURCE_PATH}")
display(Code(dashboard_html_code, language="html"))

Dashboard source loaded from: c:\Users\davie\OneDrive\Documentos\Data
Analyst\marketing_campaign_dashboard.html

<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-
scale=1.0">
```

```
  <title>Marketing Campaign Dashboard</title>
  <link rel="preconnect" href="https://fonts.googleapis.com">
  <link rel="preconnect" href="https://fonts.gstatic.com" crossorigin>
  <link href="https://fonts.googleapis.com/css2?
family=Manrope:wght@400;500;600;700;800&family=Source+Serif+4:wght@600
;700&display=swap" rel="stylesheet">
  <script src="https://cdn.plot.ly/plotly-2.35.2.min.js"></script>
  <style>
    :root {
      --ink: #1d2a3a;
      --muted: #64748b;
      --paper: #f7f3eb;
      --card: rgba(255, 252, 247, 0.88);
      --line: rgba(29, 42, 58, 0.12);
      --navy: #20364f;
      --teal: #2f7f7b;
      --gold: #ca8f2c;
      --rose: #c96b59;
      --cream: #fffdf8;
      --shadow: 0 18px 50px rgba(32, 54, 79, 0.12);
    }

    * {
      box-sizing: border-box;
    }

    body {
      margin: 0;
      font-family: 'Manrope', sans-serif;
      color: var(--ink);
      background:
        radial-gradient(circle at top left, rgba(201, 107, 89, 0.18),
transparent 28%),
        radial-gradient(circle at top right, rgba(47, 127, 123, 0.2),
transparent 26%),
        linear-gradient(180deg, #f2ede4 0%, #f8f5ef 42%, #fbfaf7
100%);
    }

    .shell {
      max-width: 1400px;
      margin: 0 auto;
      padding: 32px 24px 56px;
    }

    .hero {
      position: relative;
      overflow: hidden;
      background: linear-gradient(135deg, rgba(255,255,255,0.92),
rgba(247,243,235,0.78));
```

```
      border: 1px solid rgba(255,255,255,0.65);
      border-radius: 28px;
      box-shadow: var(--shadow);
      padding: 34px;
      display: grid;
      grid-template-columns: 1.35fr 0.9fr;
      gap: 28px;
    }

    .hero::after {
      content: "";
      position: absolute;
      inset: auto -80px -80px auto;
      width: 220px;
      height: 220px;
      border-radius: 999px;
      background: linear-gradient(135deg, rgba(202,143,44,0.25),
rgba(47,127,123,0.12));
      filter: blur(8px);
    }

    .eyebrow {
      display: inline-flex;
      align-items: center;
      gap: 10px;
      padding: 8px 14px;
      border-radius: 999px;
      background: rgba(32, 54, 79, 0.08);
      color: var(--navy);
      font-size: 12px;
      font-weight: 800;
      letter-spacing: 0.12em;
      text-transform: uppercase;
    }

    h1 {
      margin: 16px 0 14px;
      font-family: 'Source Serif 4', serif;
      font-size: clamp(2.2rem, 3.6vw, 4.1rem);
      line-height: 0.96;
      letter-spacing: -0.04em;
    }

    .hero p {
      margin: 0;
      max-width: 760px;
      font-size: 1rem;
      line-height: 1.7;
      color: rgba(29, 42, 58, 0.84);
    }
```

```
    .recommendation-card {
      align-self: stretch;
      background: linear-gradient(160deg, rgba(32,54,79,0.95),
rgba(47,127,123,0.92));
      color: white;
      border-radius: 24px;
      padding: 24px;
      display: flex;
      flex-direction: column;
      justify-content: space-between;
      box-shadow: 0 20px 40px rgba(32, 54, 79, 0.22);
    }

    .recommendation-card .label {
      font-size: 12px;
      text-transform: uppercase;
      letter-spacing: 0.14em;
      opacity: 0.78;
      font-weight: 700;
    }

    .recommendation-card h2 {
      margin: 10px 0 10px;
      font-family: 'Source Serif 4', serif;
      font-size: 2rem;
      line-height: 1.02;
      letter-spacing: -0.03em;
    }

    .recommendation-card p {
      color: rgba(255,255,255,0.84);
      line-height: 1.65;
      margin: 0 0 16px;
    }

    .impact-strip {
      display: grid;
      grid-template-columns: repeat(3, minmax(0, 1fr));
      gap: 10px;
    }

    .impact-pill {
      padding: 12px 14px;
      background: rgba(255,255,255,0.12);
      border: 1px solid rgba(255,255,255,0.16);
      border-radius: 18px;
    }

    .impact-pill strong {
```

```
      display: block;
      font-size: 1.1rem;
      margin-bottom: 4px;
    }

    .impact-pill span {
      font-size: 0.82rem;
      color: rgba(255,255,255,0.78);
    }

    .grid {
      display: grid;
      grid-template-columns: repeat(12, minmax(0, 1fr));
      gap: 20px;
      margin-top: 22px;
    }

    .panel {
      background: var(--card);
      border: 1px solid rgba(255,255,255,0.7);
      border-radius: 24px;
      box-shadow: var(--shadow);
      padding: 22px;
      backdrop-filter: blur(10px);
    }

    .span-3 { grid-column: span 3; }
    .span-4 { grid-column: span 4; }
    .span-5 { grid-column: span 5; }
    .span-6 { grid-column: span 6; }
    .span-7 { grid-column: span 7; }
    .span-8 { grid-column: span 8; }
    .span-12 { grid-column: span 12; }

    .metric-card {
      min-height: 168px;
      display: flex;
      flex-direction: column;
      justify-content: space-between;
    }

    .metric-label {
      font-size: 0.78rem;
      letter-spacing: 0.12em;
      text-transform: uppercase;
      color: var(--muted);
      font-weight: 800;
    }

    .metric-value {
```

```
      margin-top: 10px;
      font-size: clamp(1.8rem, 2.8vw, 2.7rem);
      font-weight: 800;
      letter-spacing: -0.04em;
      color: var(--navy);
    }

    .metric-note {
      color: rgba(29, 42, 58, 0.72);
      line-height: 1.6;
      font-size: 0.94rem;
    }

    .section-title {
      display: flex;
      align-items: baseline;
      justify-content: space-between;
      gap: 16px;
      margin-bottom: 10px;
    }

    .section-title h3 {
      margin: 0;
      font-size: 1.15rem;
      letter-spacing: -0.02em;
    }

    .section-title span {
      color: var(--muted);
      font-size: 0.9rem;
    }

    .chart {
      width: 100%;
      min-height: 360px;
    }

    .insight-list {
      margin: 10px 0 0;
      padding: 0;
      list-style: none;
      display: grid;
      gap: 12px;
    }

    .insight-list li {
      padding: 14px 16px;
      border-radius: 16px;
      background: rgba(32, 54, 79, 0.05);
      border: 1px solid rgba(32, 54, 79, 0.08);
```

```
      line-height: 1.6;
    }

    .insight-list strong {
      color: var(--navy);
    }

    table {
      width: 100%;
      border-collapse: collapse;
      margin-top: 14px;
      font-size: 0.94rem;
      overflow: hidden;
    }

    thead th {
      text-align: left;
      padding: 14px 12px;
      font-size: 0.78rem;
      text-transform: uppercase;
      letter-spacing: 0.08em;
      color: var(--muted);
      border-bottom: 1px solid var(--line);
    }

    tbody td {
      padding: 14px 12px;
      border-bottom: 1px solid rgba(29, 42, 58, 0.08);
    }

    tbody tr:nth-child(1) {
      background: rgba(202, 143, 44, 0.08);
    }

    .tone {
      display: inline-flex;
      padding: 7px 10px;
      border-radius: 999px;
      font-size: 0.78rem;
      font-weight: 700;
    }

    .tone-a { background: rgba(201, 107, 89, 0.12); color: #9b4939; }
    .tone-b { background: rgba(47, 127, 123, 0.12); color: #21615e; }

    .footer-note {
      margin-top: 16px;
      color: var(--muted);
      font-size: 0.9rem;
      line-height: 1.6;
```

```
    }

    @media (max-width: 1100px) {
      .hero {
        grid-template-columns: 1fr;
      }
      .span-3, .span-4, .span-5, .span-6, .span-7, .span-8 {
        grid-column: span 12;
      }
    }

    @media (max-width: 720px) {
      .shell {
        padding: 18px 14px 36px;
      }
      .hero, .panel {
        border-radius: 20px;
        padding: 18px;
      }
      .impact-strip {
        grid-template-columns: 1fr;
      }
      table {
        display: block;
        overflow-x: auto;
      }
    }
  </style>
</head>
<body>
  <div class="shell">
    <section class="hero">
      <div>
        <div class="eyebrow">New Customer Sales Decision</div>
        <h1>Marketing campaign analysis for tone, channel, and revenue
growth.</h1>
        <p>
          Campaign A uses an informal, conversational tone. Campaign B
uses a formal, corporative tone.
          Looking only at new customers, the strongest growth move is
to scale the combination that wins on
          revenue, reaches enough prospects to matter, and still
converts above the portfolio baseline.
        </p>
      </div>
      <aside class="recommendation-card">
        <div>
          <div class="label">Recommended Focus</div>
          <h2>Campaign B + Email</h2>
```

```
          <p>
            This combination is the best lever to increase sales to
new customers because it delivers the highest
            revenue at scale while staying above the new-customer
average conversion rate.
          </p>
        </div>
        <div class="impact-strip">
          <div class="impact-pill">
            <strong>$225.3K</strong>
            <span>new-customer sales</span>
          </div>
          <div class="impact-pill">
            <strong>48.18%</strong>
            <span>conversion rate</span>
          </div>
          <div class="impact-pill">
            <strong>22.79%</strong>
            <span>share of new-customer revenue</span>
          </div>
        </div>
      </aside>
    </section>

    <section class="grid">
      <article class="panel metric-card span-3">
        <div class="metric-label">New-Customer Sales</div>
        <div class="metric-value">$988.5K</div>
        <div class="metric-note">Total revenue generated from 42,597
new-customer interactions in the dataset.</div>
      </article>
      <article class="panel metric-card span-3">
        <div class="metric-label">Average Conversion</div>
        <div class="metric-value">47.86%</div>
        <div class="metric-note">Baseline conversion rate for all new-
customer traffic across campaigns and channels.</div>
      </article>
      <article class="panel metric-card span-3">
        <div class="metric-label">B Email vs A Email</div>
        <div class="metric-value">+$47.9K</div>
        <div class="metric-note">Campaign B Email generates 27.02%
more new-customer revenue than Campaign A Email.</div>
      </article>
      <article class="panel metric-card span-3">
        <div class="metric-label">Tone Winner</div>
        <div class="metric-value">Campaign B</div>
        <div class="metric-note">The formal, corporative tone wins on
total new-customer sales in every channel.</div>
      </article>
```

```
      <article class="panel span-7">
        <div class="section-title">
          <h3>Revenue by campaign and channel</h3>
          <span>Total sales from new customers</span>
        </div>
        <div id="salesByCombo" class="chart"></div>
      </article>

      <article class="panel span-5">
        <div class="section-title">
          <h3>Decision summary</h3>
          <span>What the data says</span>
        </div>
        <ul class="insight-list">
          <li><strong>Campaign B + Email</strong> is the top growth
choice because it combines the highest revenue with a conversion rate
above average.</li>
          <li><strong>Email</strong> is already the largest
acquisition channel by new-customer revenue, so improving this path
creates the biggest commercial payoff.</li>
          <li><strong>Campaign A + Instagram</strong> has the highest
conversion rate at 48.72%, but its smaller scale keeps total sales
well below B + Email.</li>
          <li><strong>Campaign B + Instagram</strong> is the best
secondary test bed because it has the strongest sales-per-lead result
at $23.48.</li>
        </ul>
      </article>

      <article class="panel span-5">
        <div class="section-title">
          <h3>Scale versus efficiency</h3>
          <span>Bubble size = lead volume</span>
        </div>
        <div id="efficiencyPlot" class="chart"></div>
      </article>

      <article class="panel span-7">
        <div class="section-title">
          <h3>Tone effect by channel</h3>
          <span>Campaign B minus Campaign A among new customers</span>
        </div>
        <div id="toneDelta" class="chart"></div>
      </article>

      <article class="panel span-12">
        <div class="section-title">
          <h3>Ranked combinations</h3>
          <span>Prioritize by sales impact first, then
```

```
efficiency</span>
        </div>
        <table>
          <thead>
            <tr>
              <th>Rank</th>
              <th>Campaign</th>
              <th>Tone</th>
              <th>Channel</th>
              <th>Leads</th>
              <th>Conversion</th>
              <th>Total Sales</th>
              <th>Sales / Lead</th>
            </tr>
          </thead>
          <tbody id="rankingBody"></tbody>
        </table>
        <p class="footer-note">
          Recommendation logic is based on revenue, conversion, and
scale. The source data does not include media spend,
          so this dashboard supports revenue optimization rather than
a full ROI decision.
        </p>
      </article>
    </section>
  </div>

  <script>
    const comboData = [
      { campaign: 'B', tone: 'Formal / corporative', channel: 'Email',
leads: 9700, conversion: 48.18, sales: 225285.52, salesPerLead: 23.23,
share: 22.79 },
      { campaign: 'A', tone: 'Informal / conversational', channel:
'Email', leads: 7683, conversion: 47.22, sales: 177366.97,
salesPerLead: 23.09, share: 17.94 },
      { campaign: 'B', tone: 'Formal / corporative', channel:
'Instagram', leads: 7100, conversion: 48.30, sales: 166722.17,
salesPerLead: 23.48, share: 16.87 },
      { campaign: 'B', tone: 'Formal / corporative', channel: 'Web
Banner', leads: 7110, conversion: 47.64, sales: 164125.79,
salesPerLead: 23.08, share: 16.60 },
      { campaign: 'A', tone: 'Informal / conversational', channel:
'Instagram', leads: 5456, conversion: 48.72, sales: 127814.93,
salesPerLead: 23.43, share: 12.93 },
      { campaign: 'A', tone: 'Informal / conversational', channel:
'Web Banner', leads: 5548, conversion: 47.12, sales: 127213.12,
salesPerLead: 22.93, share: 12.87 }
    ];

    const deltaData = [
```

```
      { channel: 'Email', salesDelta: 47918.55, salesPct: 27.02,
conversionDelta: 0.95 },
      { channel: 'Instagram', salesDelta: 38907.24, salesPct: 30.44,
conversionDelta: -0.42 },
      { channel: 'Web Banner', salesDelta: 36912.67, salesPct: 29.02,
conversionDelta: 0.52 }
    ];

    const formatCurrency = (value) => new Intl.NumberFormat('en-US', {
      style: 'currency',
      currency: 'USD',
      maximumFractionDigits: 2
    }).format(value);

    const salesByCombo = {
      data: [
        {
          x: ['Email', 'Instagram', 'Web Banner'],
          y: [177366.97, 127814.93, 127213.12],
          name: 'Campaign A',
          type: 'bar',
          marker: { color: '#c96b59', line: { color: '#9b4939', width:
1 } },
          hovertemplate: 'Campaign A<br>%{x}<br>Sales: %{y:
$,.0f}<extra></extra>'
        },
        {
          x: ['Email', 'Instagram', 'Web Banner'],
          y: [225285.52, 166722.17, 164125.79],
          name: 'Campaign B',
          type: 'bar',
          marker: { color: '#2f7f7b', line: { color: '#21615e', width:
1 } },
          hovertemplate: 'Campaign B<br>%{x}<br>Sales: %{y:
$,.0f}<extra></extra>'
        }
      ],
      layout: {
        paper_bgcolor: 'rgba(0,0,0,0)',
        plot_bgcolor: 'rgba(0,0,0,0)',
        margin: { l: 56, r: 18, t: 20, b: 48 },
        barmode: 'group',
        font: { family: 'Manrope, sans-serif', color: '#1d2a3a' },
        legend: { orientation: 'h', x: 0, y: 1.12 },
        xaxis: { tickfont: { size: 12 }, gridcolor: 'rgba(0,0,0,0)' },
        yaxis: { title: 'Sales ($)', gridcolor: 'rgba(29,42,58,0.08)',
zerolinecolor: 'rgba(29,42,58,0.08)' }
      },
      config: { responsive: true, displayModeBar: false }
```

```
    };

    Plotly.newPlot('salesByCombo', salesByCombo.data,
salesByCombo.layout, salesByCombo.config);

    Plotly.newPlot('efficiencyPlot', [{
      x: comboData.map(d => d.conversion),
      y: comboData.map(d => d.salesPerLead),
      mode: 'markers+text',
      text: comboData.map(d => `${d.campaign} ${d.channel}`),
      textposition: 'top center',
      marker: {
        size: comboData.map(d => Math.sqrt(d.leads) * 0.9),
        color: comboData.map(d => d.campaign === 'B' ? '#2f7f7b' :
'#c96b59'),
        opacity: 0.82,
        line: { color: '#ffffff', width: 1.5 }
      },
      customdata: comboData.map(d => [d.tone, d.leads, d.sales]),
      hovertemplate:
        '<b>%{text}</b><br>' +
        'Tone: %{customdata[0]}<br>' +
        'Conversion: %{x:.2f}%<br>' +
        'Sales per lead: %{y:$,.2f}<br>' +
        'Leads: %{customdata[1]:,.0f}<br>' +
        'Total sales: %{customdata[2]:$,.0f}<extra></extra>'
    }], {
      paper_bgcolor: 'rgba(0,0,0,0)',
      plot_bgcolor: 'rgba(0,0,0,0)',
      margin: { l: 58, r: 18, t: 24, b: 54 },
      font: { family: 'Manrope, sans-serif', color: '#1d2a3a' },
      xaxis: {
        title: 'Conversion rate (%)',
        gridcolor: 'rgba(29,42,58,0.08)',
        zerolinecolor: 'rgba(29,42,58,0.08)'
      },
      yaxis: {
        title: 'Sales per lead ($)',
        gridcolor: 'rgba(29,42,58,0.08)',
        zerolinecolor: 'rgba(29,42,58,0.08)'
      }
    }, { responsive: true, displayModeBar: false });

    Plotly.newPlot('toneDelta', [
      {
        x: deltaData.map(d => d.channel),
        y: deltaData.map(d => d.salesDelta),
        type: 'bar',
        name: 'Revenue lift from B vs A',
        marker: { color: '#20364f' },
```

```
        yaxis: 'y',
        hovertemplate: '%{x}<br>Revenue lift:
%{y:$,.0f}<extra></extra>'
      },
      {
        x: deltaData.map(d => d.channel),
        y: deltaData.map(d => d.conversionDelta),
        type: 'scatter',
        mode: 'lines+markers+text',
        name: 'Conversion delta (pp)',
        marker: { color: '#ca8f2c', size: 10 },
        line: { color: '#ca8f2c', width: 3 },
        text: deltaData.map(d => `${d.conversionDelta > 0 ? '+' : ''}$
{d.conversionDelta.toFixed(2)} pp`),
        textposition: 'top center',
        yaxis: 'y2',
        hovertemplate: '%{x}<br>Conversion delta: %{y:.2f}
pp<extra></extra>'
      }
    ], {
      paper_bgcolor: 'rgba(0,0,0,0)',
      plot_bgcolor: 'rgba(0,0,0,0)',
      margin: { l: 56, r: 56, t: 24, b: 48 },
      font: { family: 'Manrope, sans-serif', color: '#1d2a3a' },
      legend: { orientation: 'h', x: 0, y: 1.13 },
      xaxis: { gridcolor: 'rgba(0,0,0,0)' },
      yaxis: {
        title: 'Revenue lift ($)',
        gridcolor: 'rgba(29,42,58,0.08)',
        zerolinecolor: 'rgba(29,42,58,0.08)'
      },
      yaxis2: {
        title: 'Conversion delta (pp)',
        overlaying: 'y',
        side: 'right',
        showgrid: false,
        zerolinecolor: 'rgba(29,42,58,0.08)'
      }
    }, { responsive: true, displayModeBar: false });

    const rankingBody = document.getElementById('rankingBody');
    comboData
      .slice()
      .sort((a, b) => b.sales - a.sales)
      .forEach((row, index) => {
        const tr = document.createElement('tr');
        const toneClass = row.campaign === 'B' ? 'tone tone-b' : 'tone
tone-a';
        tr.innerHTML = `
```

```
          <td>${index + 1}</td>
          <td><strong>${row.campaign}</strong></td>
          <td><span class="${toneClass}">${row.tone}</span></td>
          <td>${row.channel}</td>
          <td>${row.leads.toLocaleString('en-US')}</td>
          <td>${row.conversion.toFixed(2)}%</td>
          <td>${formatCurrency(row.sales)}</td>
          <td>${formatCurrency(row.salesPerLead)}</td>
        `;
        rankingBody.appendChild(tr);
      });
  </script>
</body>
</html>
```

Dashboard metrics explained

The dashboard metrics are all designed to answer one business question:

Which campaign and channel combination should we focus on to increase sales to new customers?

KPI cards

1. New-Customer Sales: $988.5K
This is the total revenue generated by all new-customer interactions in the dataset. It represents the full sales opportunity being evaluated.

2. Average Conversion: 47.86%
This is the baseline conversion rate across all new-customer traffic. It is the benchmark used to judge whether a campaign and channel combination is performing above or below average.

3. B Email vs A Email: +$47.9K
This metric compares the sales generated by Campaign B Email versus Campaign A Email for new customers. It shows the incremental revenue advantage of the formal Campaign B tone in the Email channel.

4. Tone Winner: Campaign B
This is a summary conclusion from the data. It means Campaign B, which uses the more formal and corporative tone, produces more total new-customer sales than Campaign A across every channel.

Charts

1. Revenue by campaign and channel
This chart compares total sales for Campaign A and Campaign B across Email, Instagram, and Web Banner. It identifies which combination produces the most revenue. In this case, Campaign B + Email is the strongest performer.

2. Scale versus efficiency
This chart combines three performance signals:

- Conversion rate on the x-axis
- Sales per lead on the y-axis
- Lead volume through bubble size

It helps separate combinations that are efficient from combinations that are both efficient and large enough to matter commercially.

3. Tone effect by channel
This chart compares Campaign B against Campaign A by channel using:

- Revenue lift in dollars
- Conversion delta in percentage points

It shows whether the more formal tone improves performance, and by how much, in each acquisition channel.

Ranked table metrics

1. Leads
The number of new-customer interactions in that campaign and channel combination.

2. Conversion
The percentage of those leads that converted into a sale.

3. Total Sales
The total revenue produced by that combination. This is the main business outcome metric.

4. Sales / Lead
The average revenue generated per lead. This is an efficiency metric that combines conversion behavior and revenue impact.

5. Tone
This labels the communication style used in the campaign:

- Campaign A: informal and conversational
- Campaign B: formal and corporative

Why the dashboard recommends Campaign B + Email

Campaign B + Email is recommended because it performs well across the three most important dimensions:

- **Scale**: it reaches a large number of new-customer leads.
- **Conversion**: it converts above the overall new-customer average.
- **Revenue**: it generates the highest total sales of any campaign and channel combination.

Important limitation

The dataset does not include media spend or acquisition cost. That means the dashboard supports a **revenue and conversion** decision, not a full **ROI or profitability** decision.